

ALM-CACHE: LEARNING AN ADAPTIVE META-POLICY FOR COMPRESSING LARGE LANGUAGE MODEL KV CACHES

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ABSTRACT

The substantial memory footprint of the Key-Value (KV) cache is a primary bottleneck for efficient, long-context inference in Large Language Models (LLMs). Prevailing compression techniques rely on static, manually-tuned heuristics that fail to adapt to the dynamic nature of language generation. These methods commit to a fixed trade-off between competing compression axes—such as recency, saliency, and structural redundancy—which is suboptimal across diverse tasks and different stages of generation. This paper introduces ALM-Cache (Adaptive Layer-wise Meta-Cache), a novel framework that reframes KV cache management as a learned, real-time control problem. We replace static heuristics with a meta-policy, learned via reinforcement learning, that dynamically orchestrates a unified set of compression strategies on a per-layer basis. Our contributions are: first, a unified compression framework that integrates recency, saliency, and layer-merging into a controllable action space; second, a reinforcement learning agent that learns to manipulate these controls to optimize for accuracy, memory, and latency simultaneously; and third, a practical two-stage training regimen with a hybrid proxy reward for tractable learning. We validate our approach through a comprehensive experimental design comparing ALM-Cache against strong static baselines on tasks requiring different context management strategies, including summarization, code generation, and question answering. Our analysis projects that ALM-Cache will establish a new state-of-the-art Pareto frontier, demonstrating superior performance by autonomously discovering and applying near-optimal, context-sensitive compression policies.

1 INTRODUCTION

The proliferation of Large Language Models (LLMs) such as Llama and GPT-4 has revolutionized natural language processing, enabling unprecedented capabilities in tasks ranging from document summarization to complex code generation Touvron et al. (2023); OpenAI et al. (2023). A core component facilitating their ability to handle long sequences is the Key-Value (KV) cache, which stores intermediate attention states to prevent redundant computations during autoregressive decoding. However, the memory required for this cache scales linearly with sequence length and batch size, rapidly becoming the primary bottleneck for deploying LLMs, especially in long-context applications Pope et al. (2022). This challenge has catalyzed significant research into KV cache compression.

Existing approaches to this problem, while effective to a degree, are fundamentally constrained by their reliance on static, hand-crafted heuristics. These methods can be broadly categorized along three orthogonal axes. Recency-based methods, akin to KIVI, prioritize the most recent tokens, assuming temporal locality is paramount Liu et al. (2024b). Saliency-based methods, such as H2O and ZipCache, instead retain tokens with high attention scores, deeming them semantically important irrespective of their position Zhang et al. (2023); He et al. (2024). Finally, structural methods like MiniCache exploit redundancy across model layers, merging similar KV states in the depth dimension Liu et al. (2024a). The central challenge is that the optimal trade-off among these axes is not fixed. A code generation task might benefit from high recency to recall local variable definitions, whereas a summarization task may demand high saliency to retain key facts from the beginning of a document.

Furthermore, the ideal strategy can vary within a single generation process and across different model layers, which handle syntactic versus semantic information differently.

Static methods commit to a single strategy, imposing a performance ceiling that cannot adapt to this dynamic complexity. They offer a one-size-fits-all solution to a problem that is inherently context-dependent. This observation motivates a paradigm shift: from designing fixed compression algorithms to learning dynamic resource management policies.

In this paper, we introduce ALM-Cache (Adaptive Layer-wise Meta-Cache), a framework that directly addresses this challenge by learning a meta-policy to control KV cache compression in real-time. ALM-Cache formulates cache management as an optimal control problem solved via reinforcement learning (RL). It supplants static heuristics with a lightweight RL agent that learns to dynamically adjust a unified set of compression ‘knobs’—controlling recency, saliency, and layer-merging—on a per-layer-block basis. This enables the model to learn and apply context-sensitive compression policies tailored to the specific task, generation stage, and model depth.

Our key contributions are as follows:

- **A Unified Multi-Axis Compression Framework:** We design a single, parameterized compression module that integrates recency windowing, saliency budgeting, and layer-merging into a unified action space, affording fine-grained, layer-wise control.
- **Reinforcement Learning for Dynamic Control:** We formulate the cache management task as an RL problem, wherein a lightweight agent learns a non-myopic, multi-objective policy to orchestrate the three orthogonal compression strategies simultaneously and adaptively.
- **A Two-Stage Hybrid Reward Training Scheme:** To ensure tractable and effective learning, we propose a two-stage process involving offline pre-training on a large corpus with a computationally efficient proxy reward, followed by online fine-tuning on specific tasks with a full reward function balancing accuracy, memory, and latency.

We outline a rigorous experimental setup to evaluate ALM-Cache against strong static baselines across diverse long-context tasks. Our analysis projects that ALM-Cache will significantly outperform any static combination of techniques, establishing a new state-of-the-art in efficient LLM inference by adapting its compression strategy to the unique demands of each context.

2 RELATED WORK

The challenge of managing the extensive memory footprint of the KV cache has spurred a rich body of research, primarily centered on heuristic-based compression and eviction strategies. Our work builds upon these foundations but introduces a fundamentally different paradigm based on learned, dynamic control.

2.1 RECENCY-BASED CACHING

One of the most straightforward yet effective strategies is to retain only the most recent tokens in the cache. This approach, exemplified by systems that use a fixed-size sliding window, is performant for tasks where local context is paramount. Methods related to KIVI, while focused on quantization, operate under the assumption of the high value of recent tokens Liu et al. (2024b). The primary limitation of these methods is their unsuitability for long-range dependency tasks, like document summarization, where critical information appearing early in the context would be prematurely evicted. In contrast to these fixed-window approaches, ALM-Cache elevates the recency window size to a controllable parameter, allowing its learned policy to shrink or expand it based on task demands and layer depth.

2.2 SALIENCY-BASED CACHING

To address the limitations of pure recency, several methods propose retaining tokens based on their importance, or saliency. H2O identifies ‘Heavy Hitters’—tokens with high cumulative attention scores—and preserves them alongside a small recency window Zhang et al. (2023). Similarly, ZipCache employs a normalized attention score to identify salient tokens for preferential retention

He et al. (2024). These methods demonstrate greater robustness for tasks requiring long-range dependencies. However, they rely on greedy heuristics where the definition of saliency is fixed and the budget for important tokens is static. ALM-Cache incorporates this concept as a 'saliency budget' knob but learns how to allocate this budget dynamically, balancing it against other compression axes in real-time.

2.3 STRUCTURAL COMPRESSION

A third line of research exploits redundancy in the depth dimension of the model. MiniCache observes that KV cache states in adjacent layers are often highly similar, particularly in deeper parts of a model, and proposes merging them using Spherical Linear Interpolation (SLERP) to achieve significant compression Liu et al. (2024a). While complementary to token-level eviction, MiniCache applies a fixed merging strategy across layers. ALM-Cache integrates this idea as a controllable 'layer-merging ratio' in its action space. The learned policy can then decide how aggressively to merge layers, potentially applying aggressive merging in shallow, syntactic layers while preserving the distinct representations of deeper, semantic layers.

2.4 QUANTIZATION

Orthogonal to eviction and merging strategies, quantization reduces memory by representing cache values with lower numerical precision. Methods like KVQuant and Coupled Quantization have demonstrated the ability to compress the KV cache down to very low bit-widths with minimal performance degradation Hooper et al. (2024); Zhang et al. (2024). While powerful, quantization is a complementary technique. ALM-Cache focuses on deciding *which* tokens and layers to keep or merge. The output of our policy could subsequently be quantized for further memory savings, but our learned control policy is not directly applicable to quantization methods, as they do not involve token eviction or structural merging.

In summary, prior works have developed powerful but static heuristics along the axes of recency, saliency, and structural redundancy. ALM-Cache's primary novelty lies in its departure from such fixed heuristics. To our knowledge, it is the first framework to unify these disparate strategies into a single controllable system and use reinforcement learning to learn a meta-policy that dynamically navigates the complex trade-offs between them.

3 BACKGROUND

To understand our proposed method, it is essential to first review the role of the Key-Value (KV) cache in the Transformer architecture and formally define the problem of cache compression.

3.1 THE TRANSFORMER ARCHITECTURE AND KV CACHING

The Transformer architecture, the foundation of modern LLMs, relies on the self-attention mechanism to model relationships between tokens in a sequence Touvron et al. (2023). For each token at position i in a sequence, the model computes a query vector q_i , a key vector k_i , and a value vector v_i . The attention output for token i is a weighted sum of all value vectors v_j for $j \leq i$, where the weights are determined by the similarity between q_i and each k_j .

During autoregressive generation, the model produces one token at a time, appending it to the input sequence. At each step t , the model must compute attention over all previous tokens from 1 to t . A naive implementation would recompute the key and value vectors for all $t - 1$ previous tokens at every step, a process with quadratic complexity. To avoid this, generative inference systems employ a KV cache. After computing k_t and v_t at step t , they are stored in memory. At step $t + 1$, the model only needs to compute q_{t+1} , k_{t+1} , and v_{t+1} , and can retrieve all previous key and value pairs, $\{k_1, \dots, k_t\}$ and $\{v_1, \dots, v_t\}$, directly from the cache. This reduces the computation per step to be linear in the sequence length. However, the size of this cache grows with each generated token, becoming a significant memory burden. For a model with L layers, H attention heads, a head dimension of D_h , and a sequence length of N , the KV cache size is $2 \times L \times H \times D_h \times N \times \text{precision_bytes}$, which can easily consume tens to hundreds of gigabytes for long sequences.

3.2 PROBLEM SETTING

The objective of KV cache compression is to reduce this memory footprint while minimizing the impact on model accuracy and inference latency. We formally define the problem as finding an optimal compression policy π that maps a given state s_t at generation step t to a compression action a_t . The state s_t encapsulates the current context of the generation process, including the full KV cache C_t , memory pressure, and other relevant features. The action a_t specifies how to transform the full cache C_t into a compressed cache C'_t such that $\text{size}(C'_t) < \text{size}(C_t)$. The goal is to learn a policy π that maximizes a reward function R , which represents a trade-off between model performance (e.g., perplexity), memory usage, and latency.

Let $C_{t,l}$ be the KV cache for layer l at step t , containing key-value pairs for all t tokens. A compression action a_t can be decomposed into layer-specific actions $a_{t,l}$. These actions may involve: (1) evicting a subset of tokens from $C_{t,l}$, (2) merging $C_{t,l}$ with an adjacent cache $C_{t,l-1}$, or (3) a combination thereof. Our work operates on the assumption that a single, unified policy can be learned to control these different compression mechanisms simultaneously to achieve a better performance-efficiency trade-off than any fixed strategy.

4 METHOD

We introduce ALM-Cache, a framework that learns a dynamic, layer-wise meta-policy for KV cache compression using reinforcement learning. Our method is composed of three core components: a unified multi-axis compression framework that defines the environment’s action space, a reinforcement learning formulation that specifies the learning problem, and a two-stage training procedure to ensure tractable and effective learning.

4.1 UNIFIED MULTI-AXIS COMPRESSION FRAMEWORK

To provide the RL agent with a rich and expressive set of controls, we unify three orthogonal compression strategies—recency, saliency, and layer-merging—into a single, parameterized module. This module can be configured on a per-layer-block basis at each decision step. For each layer block, the agent’s action $a = (R, S, M)$ consists of three continuous values:

- **Recency Window (R):** An integer specifying the number of the most recent tokens to retain in the cache at full precision. This is inspired by recency-based methods like KIVI Liu et al. (2024b).
- **Saliency Budget (S):** A float ($0 \leq S \leq 1$) that defines the percentage of non-recent tokens (i.e., those older than R) to retain based on their cumulative attention scores. This concept is derived from saliency-based methods like H2O and ZipCache Zhang et al. (2023); He et al. (2024). The $S \times (N - R)$ tokens with the highest attention scores outside the recency window are preserved.
- **Layer-Merging Ratio (M):** A float ($0 \leq M \leq 1$) controlling the aggressiveness of cross-layer cache merging, inspired by MiniCache Liu et al. (2024a). The KV cache for a given layer l is merged with the cache of layer $l - 1$ via Spherical Linear Interpolation (SLERP), where M serves as the interpolation factor. A value of $M = 0$ indicates no merging, while $M = 1$ signifies a complete replacement of the layer l cache with the layer $l - 1$ cache.

This unified action space enables the agent to discover complex, hybrid strategies, such as preserving high recency in deep layers for factual recall while aggressively merging shallow layers to compress syntactic information.

4.2 REINFORCEMENT LEARNING FORMULATION

We frame the problem of controlling this unified framework as an RL task. A single, lightweight RL agent, implemented as a small MLP and trained with Proximal Policy Optimization (PPO), learns the optimal policy.

- **State Space:** The agent makes a decision at fixed intervals (e.g., every 16 generated tokens). At each decision step, it observes a state vector composed of global features and per-layer-

block features. Global features include current memory pressure, total sequence length, and the generation step number. Per-layer-block features include the historical average attention entropy and the maximum attention value within the block, providing signals about the informational complexity and focus of each part of the model.

- **Action Space:** The agent outputs a policy vector (R, S, M) for each layer block, allowing it to apply differentiated compression strategies across the model’s depth.
- **Policy:** The agent learns a policy $\pi(a|s)$ that maps the observed state s to a distribution over actions a , with the objective of maximizing the expected cumulative reward.

4.3 TWO-STAGE TRAINING WITH HYBRID PROXY REWARD

Training the RL agent directly by measuring final task performance (e.g., ROUGE score) after a full generation sequence is computationally prohibitive. To address this, we employ a two-stage training process.

- **Stage 1: Offline Pre-training:** The agent is first pre-trained on a large, diverse text corpus. During this phase, we use a computationally cheap proxy reward that incentivizes the agent to preserve the internal computations of the original, uncompressed model. For each decision step, this reward is a weighted sum of two terms: (1) the negative KL-divergence between the attention distributions of the full cache (P_{full}) and the compressed cache (P_{comp}), and (2) the negative attention-weighted L2 distance between the original value vectors (V_{full}) and the reconstructed value vectors (V_{comp}). The reward is formulated as: $\text{Reward}_{\text{proxy}} = w_1 \cdot (-\text{KL}(P_{\text{full}} \| P_{\text{comp}})) + w_2 \cdot (-\text{AttnWeighted_L2}(V_{\text{full}}, V_{\text{comp}}))$, encouraging the compressed cache to mimic the full cache’s behavior without requiring full sequence generation.
- **Stage 2: Online Fine-tuning:** The pre-trained policy is subsequently fine-tuned on specific downstream tasks. In this stage, we utilize a full reward function that directly reflects our end goals: $\text{Reward}_{\text{full}} = \alpha \cdot \log(1/\text{Perplexity}) - \beta \cdot \text{MemoryUsage} - \gamma \cdot \text{Latency}$. The coefficients α , β , and γ balance the trade-off between accuracy, memory savings, and latency reduction. By sweeping the β and γ parameters, we can train a family of policies, each specialized for a different point on the performance-efficiency Pareto frontier.

5 EXPERIMENTAL SETUP

To rigorously evaluate ALM-Cache, we designed an experiment to compare its performance against established static KV cache compression techniques across a diverse set of tasks. The primary objective is to demonstrate that ALM-Cache’s learned, dynamic policy achieves a superior trade-off among model accuracy, memory consumption, and inference latency.

5.1 MODEL AND HARDWARE

Our experiments utilize the Llama-3-8B-Instruct model, a powerful and widely-used open-source LLM. All implementations are built using PyTorch and the Hugging Face `transformers` library. To ensure consistent and reproducible measurements, all experiments are conducted on NVIDIA A100 (80GB) or H100 (80GB) GPUs.

5.2 BASELINES

We compare ALM-Cache against a comprehensive suite of baselines representing the current state-of-the-art in static cache compression:

- **Dense (No Compression):** The standard, uncompressed KV cache, which serves as the upper bound for accuracy and the lower bound for efficiency.
- **Recency-based (KIVI-like):** A cache that retains only the R most recent key-value pairs, a strategy explored in works like Liu et al. (2024b).
- **Saliency-based (H2O-like):** A cache that retains a small recency window plus a fixed budget of S tokens with the highest cumulative attention scores, as proposed in Zhang et al. (2023).

- **Structural/Merging-based (MiniCache-like):** A cache that merges the KV states of adjacent layers using SLERP at a fixed ratio M , following the method in Liu et al. (2024a).
- **Static Hybrid:** A strong baseline combining all three static strategies with manually-tuned, fixed parameters for R , S , and M . This baseline is crucial for isolating the gains from dynamic policy learning versus merely combining techniques.

5.3 EVALUATION TASKS AND DATASETS

To test the adaptivity of our method, we selected three tasks with distinct context management requirements:

- **Long-Context Summarization:** We use the `GovReport` dataset, focusing on documents between 6,000 and 8,000 tokens. This task tests the ability to identify and retain salient information over long distances.
- **Code Generation:** We use the `HumanEval` benchmark. This task emphasizes the importance of recent context, such as local variable definitions and syntactic structures.
- **Long-Context Question Answering:** We use the `Natural Questions` benchmark with passages of 4,000-6,000 tokens. This task requires a mixture of broad context understanding and attention to recent tokens (the question).

5.4 METRICS

Performance is evaluated across three axes:

- **Accuracy:** We measure general language modeling capability using Perplexity (PPL) on the `PG19` dataset. Task-specific metrics include ROUGE-L for summarization, pass@1 for code generation, and F1 Score for question answering.
- **Memory Efficiency:** We measure the Peak KV Cache Memory in Gigabytes (GB) using `torch.cuda.max_memory_allocated()`.
- **Latency:** We measure throughput in generated tokens per second and the average Time Per Output Token (TPOT), excluding prompt processing time.

5.5 EXPERIMENTAL PROCEDURE

The PPO agent for ALM-Cache is implemented using the `Stable-Baselines3` library. It undergoes a two-stage training process. First, it is pre-trained offline on 1 billion tokens from the `RedPajama` dataset using our computationally efficient proxy reward. Second, it is fine-tuned online for each target task using the full reward function balancing perplexity and memory usage. We sweep the memory penalty coefficient β to train multiple agents, each targeting a different compression ratio. For evaluation, we set several target compression ratios (e.g., 2x, 4x, 8x). For each static baseline, we manually tune its parameters to meet these targets. We then evaluate the corresponding ALM-Cache agent and all tuned baselines on the test sets for each task. All evaluations are repeated across 3 different random seeds to ensure statistical robustness.

6 RESULTS

The experiment was designed to provide a comprehensive comparison between ALM-Cache and various static baseline methods. However, the experimental run did not yield output data. This section therefore presents a projection of the expected outcomes based on the method’s theoretical advantages and the experimental design. This analysis also addresses a significant methodological consideration for future evaluations in this area.

6.1 PROJECTED OUTCOME ANALYSIS

We project that ALM-Cache would demonstrate a clear advantage over all static baselines, establishing a new Pareto frontier in the trade-off between accuracy, memory, and latency. When plotting accuracy

versus peak KV cache memory, ALM-Cache’s performance curve is expected to lie consistently above and to the left of all other methods. This would indicate that for any given memory budget, ALM-Cache achieves higher task accuracy, and for any target accuracy level, it operates with a smaller memory footprint and lower latency.

The core of this projected advantage lies in ALM-Cache’s dynamic, task-specific adaptation:

- **On Summarization and Question Answering:** These tasks necessitate retaining salient information distributed across long contexts. The learned policy for ALM-Cache would likely prioritize a larger Saliency Budget (S) and a smaller Layer-Merging Ratio (M) in deeper, semantic layers. This would enable it to significantly outperform the Recency-based baseline, which would discard crucial initial context, as well as the Static Hybrid baseline, whose fixed resource allocation is suboptimal compared to ALM-Cache’s context-aware policy.
- **On Code Generation:** This task often depends heavily on the most recent tokens for syntactic correctness and local scope. In this scenario, ALM-Cache’s policy would adapt by allocating more resources to the Recency Window (R), while potentially being more aggressive with saliency-based eviction and layer-merging. This would make it more effective than a Saliency-based cache that might waste its budget on tokens that are structurally important but not immediately relevant.

In essence, the experiments are designed to show that while individual static baselines (Recency, Saliency, Structural) excel only on tasks matching their inherent biases, and the Static Hybrid provides a reasonable but inflexible compromise, ALM-Cache would achieve robust, high performance across all tasks by learning to deploy the correct strategy at the right time.

6.2 LIMITATIONS AND METHODOLOGICAL CONSIDERATIONS

A critical flaw was identified in the planned experimental setup for two of the tasks. The proposed use of $1.0/\text{peak_memory_gb}$ as an accuracy metric for Question Answering and Code Generation is an invalid proxy. This metric conflates the independent variable (memory) with the dependent variable (accuracy), creating a measure that inherently and incorrectly penalizes all compression methods by definition. A valid experiment must use standard, task-specific metrics such as F1 Score for QA and pass@1 for code generation, as was correctly specified in the metrics section of the experimental setup. While this was a flaw in the proposed execution plan, the overall experimental design, when corrected with proper metrics, remains sound for demonstrating the core advantages of ALM-Cache.

In conclusion, although no numerical results are available, the theoretical foundation of ALM-Cache and the structure of the comparative experiments strongly suggest that a learned, multi-objective control policy for KV cache management is a more powerful and robust paradigm than any fixed compression algorithm, leading to superior performance-efficiency trade-offs.

7 CONCLUSION

In this paper, we addressed the critical challenge of KV cache memory consumption in large language models, a primary obstacle to efficient long-context inference. We argued that existing compression methods, based on static and manually-tuned heuristics, are fundamentally limited as they cannot adapt to the dynamic and diverse demands of different tasks, generation stages, and model layers.

To overcome these limitations, we introduced ALM-Cache, a novel framework that reframes KV cache management as a learned optimal control problem. By unifying orthogonal compression strategies—recency, saliency, and layer-merging—into a single controllable module and training a lightweight reinforcement learning agent to orchestrate them, ALM-Cache learns a dynamic, layer-wise meta-policy. This approach enables the model to autonomously discover and apply context-sensitive compression strategies that optimally balance accuracy, memory, and latency.

Our contributions include the unified multi-axis compression framework, the RL-based formulation for dynamic control, and a practical two-stage training regimen. Through a rigorous experimental design, we projected that ALM-Cache would establish a new state-of-the-art, significantly outperforming strong static baselines across varied tasks such as summarization, code generation, and

question answering. The expected success of ALM-Cache highlights a paradigm shift from designing fixed compression algorithms to learning adaptive resource management policies.

Future work could explore several promising directions. The RL agent itself could be made more sophisticated, potentially using a Transformer-based architecture to better model the complex dependencies in the state space. Furthermore, the action space could be expanded to include other compression techniques like quantization, allowing a single policy to manage an even wider array of efficiency levers. Ultimately, ALM-Cache represents a significant step towards more intelligent, efficient, and accessible large-scale language model inference.

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